

CLASS – IX

CHAPTER – 2

Shaping of the Earth's Surface

Section A: Multiple Choice Questions (1 Mark Each)

- Which process breaks rocks into smaller pieces without moving them?
A. Erosion B. Deposition C. **Weathering** D. Transportation
- Which agent mainly forms sand dunes?
A. River B. **Wind** C. Glacier D. Sea waves
- U-shaped valleys are formed by:
A. Rivers B. Wind C. **Glaciers** D. Rainwater
- Which landform is created by river deposition?
A. **Delta** B. Gorge C. Canyon D. Cliff
- Which process carries weathered materials from one place to another?
A. Weathering B. **Transportation** C. Folding D. Volcanism
- Sea waves mainly erode:
A. Mountain peaks B. **Coastal areas** C. Desert interiors D. Plateaus
- Which factor accelerates weathering?
A. Constant temperature B. Heavy vegetation
C. **Frequent heating and cooling** D. Lack of moisture
- Which landform is NOT formed by rivers?
A. Floodplain B. Meander C. Delta D. **Moraine**
- Which process builds new landforms?
A. Erosion B. Weathering C. **Deposition** D. Earthquake
- Which agent is most effective in deserts?
A. Glacier B. **Wind** C. River D. Ocean currents

Section B: Very Short Answer Questions (2 Marks)

11. Define weathering.

Answer:

Weathering is the process by which rocks are broken down into smaller pieces **at or near the Earth's surface** due to physical, chemical, or biological processes without being moved from their original place.

12. Differentiate between weathering and erosion.

Weathering	Erosion
It is the breaking down of rocks in their original place.	It is the wearing away and removal of weathered materials from one place to another.
No movement of rock material takes place.	Rock materials are transported by agents like rivers, wind, glaciers, and sea waves.
It is caused by physical, chemical, or biological processes.	It is caused by natural agents such as water, wind, glaciers, and waves.

13. Name any two agents of erosion.

Answer:

Any two agents of erosion are:

- Rivers (running water)
- Wind

Other agents include glaciers and sea waves.

14. What is deposition?

Answer:

Deposition is the process by which eroded and transported materials such as sand, silt, gravel, and rocks are laid down in a new location when the transporting agent loses its energy.

15. What is mechanical weathering?

Answer:

Mechanical weathering, also called **physical weathering**, is the breaking down of rocks into smaller pieces **without changing their chemical composition**. It is caused by factors such as changes in temperature, freezing and thawing of water, and plant roots growing into rock cracks.

16. Why do rivers deposit sediments near their mouths?

Answer:

As a river approaches its mouth, it enters a lake or sea where its speed decreases. The river loses the energy needed to carry sediments, causing them to settle and form depositional features such as **deltas** and **floodplains**.

17. Name two landforms formed by glaciers.

Answer:

Two landforms formed by glaciers are:

- **U-shaped valleys**
- **Moraines**

Other examples include cirques, fjords, and hanging valleys.

18. What is a floodplain?

Answer:

A floodplain is a **broad, flat area** along the banks of a river formed by the deposition of fine sediments during floods. These areas are highly fertile and suitable for agriculture.

19. Why are deltas fertile?

Answer:

Deltas are fertile because rivers deposit **nutrient-rich alluvial soil** at their mouths. This soil contains minerals and organic matter that make it ideal for growing crops.

20. What is the role of gravity in mass movement?

Answer:

Gravity is the main force responsible for **mass movement**, as it pulls weathered rocks, soil, and debris downhill. This results in processes such as **landslides, rockfalls, mudflows, and soil creep**, especially on steep slopes.

Section C: Short Answer Questions (3 Marks)

21. Explain three types of weathering.

Answer:

Weathering is of three main types:

i. Mechanical (Physical) Weathering:

- Rocks break into smaller pieces without any change in their chemical composition.
- It is caused by temperature changes, freezing and thawing, and pressure release.

ii. Chemical Weathering:

- Rocks decompose due to chemical reactions with water, oxygen, and carbon dioxide.
- It changes the mineral composition of rocks and is common in warm and humid regions.

iii. Biological Weathering:

- Rocks are broken down by the activities of plants, animals, and microorganisms.
 - Plant roots grow into rock cracks, while burrowing animals loosen the soil and rocks.
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22. Describe the work of rivers in shaping the Earth's surface.

Answer:

Rivers shape the Earth's surface through three main processes:

- **Erosion:** Rivers wear away rocks and soil, forming V-shaped valleys, gorges, and waterfalls.
- **Transportation:** Rivers carry eroded materials such as sand, pebbles, and silt downstream.
- **Deposition:** When the river slows down, it deposits sediments, forming floodplains, meanders, natural levees, and deltas.

Thus, rivers continuously reshape landscapes from their source to their mouth.

23. Explain how glaciers modify landscapes.

Answer:

Glaciers are slow-moving masses of ice that reshape the land by:

- **Eroding** rocks through plucking and abrasion.
- **Carrying** rock fragments over long distances.
- **Depositing** materials when they melt, forming moraines and drumlins.
- Creating distinctive landforms such as **U-shaped valleys, cirques, hanging valleys, and fjords.**

24. Describe the role of wind in erosion and deposition.

Answer:

Wind is an important agent of erosion in dry and desert regions.

- It **erodes** rocks by lifting and carrying fine sand particles.
- Wind-blown sand wears away rock surfaces through abrasion.
- When wind loses speed, it **deposits** sand and dust.
- This leads to the formation of **sand dunes** and **loess plains**.

25. Differentiate between erosion and deposition with examples.

Erosion	Deposition
It is the wearing away and removal of rocks and soil.	It is the settling or laying down of transported materials.
It occurs when agents have high energy.	It occurs when agents lose energy.
It destroys or wears away landforms.	It builds new landforms.
Example: Formation of gorges and valleys by rivers.	Example: Formation of deltas and floodplains by rivers.

26. Explain how sea waves shape coastal landforms.

Answer:

Sea waves constantly erode and deposit materials along coastlines.

- Strong waves erode rocks to form **sea cliffs, caves, arches, and stacks.**
- Waves transport sand and pebbles along the shore.
- When wave energy decreases, sediments are deposited, forming **beaches, spits, and sandbars.**
- Thus, sea waves continuously modify coastal landscapes.

27. Why are mountains more prone to landslides?

Answer:

Mountains are more prone to landslides because:

- They have **steep slopes**, making rocks and soil unstable.
- Heavy rainfall weakens the soil and increases its weight.
- Earthquakes can trigger sudden slope failures.
- Human activities such as deforestation, mining, and road construction further reduce slope stability.

28. Explain the formation of waterfalls.

Answer:

Waterfalls are formed when a river flows over layers of hard and soft rocks.

- The softer rock is eroded more quickly than the harder rock.
- This creates a steep vertical drop.
- Water plunges over the edge, forming a waterfall.
- Continuous erosion at the base causes the waterfall to retreat upstream over time.

29. How are meanders formed?**Answer:**

Meanders are bends or loops in a river channel.

- As a river flows across a plain, it erodes the outer bank where the water flows faster.
- It deposits sediments on the inner bank where the water flows more slowly.
- Over time, repeated erosion and deposition make the bends larger, forming meanders.

30. Explain the importance of weathering in soil formation.**Answer:**

Weathering plays a vital role in soil formation because:

- It breaks rocks into small particles that become the parent material of soil.
- It releases minerals essential for plant growth.
- It mixes with organic matter to produce fertile soil.
- Different types of weathering create different kinds of soils, supporting agriculture and natural vegetation.

Therefore, weathering is the first and most important step in the formation of soil.

31. Differentiate between the three types of plate boundaries with one example each.

Plate Boundary	Movement	Landforms Formed	Example
Divergent Boundary	Plates move apart	Mid-ocean ridges, Rift valleys	Mid-Atlantic Ridge
Convergent Boundary	Plates move towards each other	Fold mountains, Deep ocean trenches	Himalayas
Transform Boundary	Plates slide past each other	Faults, Earthquakes	San Andreas Fault

32. Why is the mantle considered the most important layer in Plate Tectonic Theory?**Answer:**

The mantle is important because:

- It contains semi-molten rocks called the asthenosphere.
- Convection currents within the mantle move tectonic plates.
- Plate movement leads to earthquakes, volcanic eruptions, and mountain building.

33. Explain how convergent plate boundaries lead to the formation of the Himalayas.**Answer:**

The Himalayas were formed when:

- The Indian Plate moved northwards.
- It collided with the Eurasian Plate.
- Neither plate sank because both are continental plates.
- The rocks were compressed, folded, and uplifted.
- This process formed the young fold mountains known as the Himalayas.

34. Why are landslides more common in mountainous regions? Suggest any three mitigation measures.**Answer:**

Reasons:

(i) Steep slopes (ii) Heavy rainfall (iii) Earthquakes (iv) Deforestation and road construction

Mitigation Measures:

1. Plant more trees to bind the soil.
 2. Construct retaining walls on slopes.
 3. Avoid indiscriminate mining and cutting of hills.
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35. Name the major plate tectonics found on Earth?

Answer:

There are 7 major plate tectonics on Earth are:-

- (i) Eurasian Plate
 - (ii) Indo – Australian Plate
 - (iii) North American Plate
 - (iv) South American Plate
 - (v) African Plate
 - (vi) Antarctic Plate
 - (vii) Pacific Plate
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36. Draw and label the interior of the Earth and discontinuity layer. Explain why scientists know about the Earth's interior even though they have never reached the core.

Section D: Long Answer Questions (5 Marks)

37. Describe the three stages of river action—erosion, transportation, and deposition—with suitable examples.

Answer:

Rivers are powerful agents that continuously shape the Earth's surface through three main processes: **erosion, transportation, and deposition.**

1. Erosion

- Erosion is the process by which a river wears away rocks, soil, and other materials from its bed and banks.
- It is most active in the upper course of the river where the slope is steep and the water flows rapidly.
- **Examples of erosional landforms:** V-shaped valleys, gorges, waterfalls, and canyons.

2. Transportation

- After erosion, the river carries the eroded materials downstream.
- The materials are transported in the form of dissolved minerals, suspended silt, rolling pebbles, and larger rocks.
- Transportation connects the upper, middle, and lower courses of the river.

3. Deposition

- When the river reaches flatter land or enters a lake or sea, its speed decreases.
- It loses the energy needed to carry sediments, which are deposited.
- **Examples of depositional landforms:** Floodplains, natural levees, meanders, oxbow lakes, and deltas.

Conclusion:

Through erosion, transportation, and deposition, rivers continuously wear down old landforms and create new ones, making them one of the most important agents shaping the Earth's surface.

38. Explain the formation of different landforms by glaciers.

Answer:

Glaciers are large masses of ice that move slowly downhill under the force of gravity. They shape the Earth's surface through erosion and deposition.

Landforms formed by glacial erosion

- **U-shaped valleys:** Formed when glaciers widen and deepen river valleys.
- **Cirques:** Bowl-shaped hollows formed at the head of glaciers.
- **Hanging valleys:** Smaller valleys left at a higher level than the main valley after glaciers melt.
- **Fjords:** Deep U-shaped valleys flooded by seawater after glaciers retreat.

Landforms formed by glacial deposition

- **Moraines:** Ridges of rocks, sand, and soil deposited by glaciers.
- **Drumlins:** Smooth, elongated hills formed by deposited glacial till.
- **Eskers:** Long winding ridges formed by sediments deposited by streams flowing beneath glaciers.

Conclusion:

Glaciers transform mountainous regions by eroding rocks and depositing sediments, creating unique and spectacular landscapes.

39. Describe the role of wind in shaping desert landscapes.**Answer:**

Wind is the most important agent of erosion and deposition in arid and semi-arid regions.

Role in erosion

- Wind picks up loose sand and dust particles.
- It erodes rocks through **deflation** (removal of loose particles) and **abrasion** (sand particles rubbing against rocks).
- These processes produce landforms such as **mushroom rocks, yardangs, and ventifacts**.

Role in transportation

- Wind carries fine particles over long distances by suspension, saltation, and surface creep.

Role in deposition

- When wind speed decreases, it deposits sand and dust.
- Deposited sand forms **sand dunes**, while fine dust forms **loess plains**.

Conclusion:

Wind continuously erodes, transports, and deposits materials, creating the distinctive landscapes found in desert regions.

40. Explain how internal and external forces together shape the Earth's surface.**Answer:**

The Earth's surface is continuously shaped by the combined action of **internal (endogenic)** and **external (exogenic)** forces.

Internal forces

- Originate from within the Earth.
- Build and uplift landforms.
- Include earthquakes, volcanic eruptions, folding, and faulting.
- They create mountains, plateaus, volcanoes, and rift valleys.

External forces

- Operate on the Earth's surface.
- Wear down and reshape landforms.
- Include weathering, erosion, transportation, and deposition by rivers, glaciers, wind, sea waves, and gravity.

Combined action

- Internal forces build high landforms.
- External forces gradually wear them down and redistribute sediments.
- Over millions of years, these two forces work together to create and modify landscapes.

Conclusion:

The present-day relief of the Earth is the result of the continuous interaction between internal forces that build landforms and external forces that wear them down.

41. Discuss the importance of weathering in agriculture and human life.**Answer:**

Weathering plays an important role in supporting agriculture, ecosystems, and human activities.

Importance in agriculture

- Breaks rocks into small particles to form soil.
- Releases essential minerals and nutrients required for plant growth.
- Improves soil fertility and water-holding capacity.
- Supports the growth of crops and vegetation.

Importance in human life

- Provides materials such as sand, clay, and gravel used in construction.
- Helps in the formation of groundwater by increasing rock permeability.
- Contributes to the formation of valuable mineral deposits.

- Creates landscapes that support tourism and biodiversity.

Environmental importance

- Helps recycle nutrients in ecosystems.
- Supports natural vegetation and wildlife habitats.

Conclusion:

Weathering is a natural process that is essential for soil formation, agriculture, construction, water storage, and maintaining ecological balance, making it vital for both nature and human life.

42. Explain the Plate Tectonic Theory. How does it help us understand the shaping of the Earth's surface?

Answer:

The **Plate Tectonic Theory** states that the Earth's lithosphere is divided into several rigid plates that float over the semi-molten asthenosphere. These plates move slowly due to convection currents in the mantle.

The movement of plates results in:

- Formation of mountains
- Earthquakes
- Volcanic eruptions
- Ocean basins and rift valleys

Thus, the movement of tectonic plates continuously reshapes the Earth's surface.

43. Explain the role of early warning systems in disaster management.

Answer:

Early warning systems:

- Detect hazards such as cyclones, floods, and tsunamis.
 - Provide timely information to authorities and the public.
 - Help in evacuation and rescue planning.
 - Reduce casualties and property damage.
 - Improve community preparedness.
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44. Explain how community participation plays an important role in disaster management. (5 Marks)

Answer:

Community participation is essential because local people are the first responders during disasters.

They can:

- Participate in awareness campaigns.
- Conduct mock drills.
- Prepare emergency kits.
- Assist in evacuation and rescue operations.
- Support relief and rehabilitation efforts.
- Share accurate information and help vulnerable groups.

An informed and prepared community significantly reduces disaster risks and strengthens resilience.

45. A new highway is being constructed through a mountainous region. Which natural hazard related to mass movement should engineers consider? Suggest two preventive measures.

Answer:

The major natural hazard is **landslides**.

Landslides occur when rocks, soil, and debris move rapidly down steep slopes due to gravity. Heavy rainfall, earthquakes, and human activities such as road cutting can trigger landslides.

Two preventive measures are:

1. **Construct retaining walls and proper drainage systems** to support slopes and remove excess rainwater.
2. **Plant trees and avoid unnecessary cutting of hill slopes**, as tree roots help bind the soil and reduce slope instability.

Conclusion:

Careful planning and proper engineering techniques can significantly reduce the risk of landslides during highway construction.

46. Explain how rivers can act both as destructive and constructive agents in shaping the Earth's surface.**Answer:**

Rivers continuously shape the Earth's surface through the processes of **erosion, transportation, and deposition**.

Rivers as Destructive Agents

- Rivers erode rocks and soil, especially in their upper course where the slope is steep.
- This erosion forms landforms such as **V-shaped valleys, gorges, waterfalls, and canyons**.
- Continuous erosion changes the landscape over time.

Rivers as Constructive Agents

- Rivers transport sediments and deposit them when their speed decreases.
- Deposition forms fertile **floodplains, natural levees, meanders, oxbow lakes, and deltas**.
- These landforms support agriculture, human settlements, and biodiversity.

Conclusion:

Rivers are **destructive** because they wear away the Earth's surface through erosion, and **constructive** because they build new landforms through deposition. Together, these processes continuously reshape the Earth's landscape.

47.

Assertion (A): Weathering does not involve movement of rock material.

Reason (R): Weathering occurs at the place where rocks are found.

A. Both A and R are true and R is the correct explanation.

B. Both A and R are true but R is not the correct explanation.

C. A is true but R is false.

D. A is false but R is true.

48.

Assertion: Rivers form deltas near their mouths.

Reason: River velocity increases before entering the sea.

A. Both A and R are true and R is the correct explanation.

B. Both A and R are true but R is not the correct explanation.

C. A is true but R is false.

D. A is false but R is true.

49.

Assertion: Glaciers are powerful agents of erosion.

Reason: Moving ice carries rocks that scrape the land.

A. Both true and R explains A

B. Both true but R does not explain A

C. A true, R false

D. A false, R true

50.

Assertion: Water erosion is most active in deserts.

Reason: Deserts have little vegetation to hold the soil.

A. Both true and R explains A

B. Both true but R does not explain A

C. A true, R false

D. A false, R true

51.

Assertion: Erosion occurs when transporting agents carry soil.

Reason: Materials settle when they can no longer be carried.

A. Both true and R explains A

B. Both true but R does not explain A

C. A true, R false

D. A false, R true

Section F: Case-Based Questions

Case 1

Read the passage carefully.

A river begins in the mountains where the slope is steep. It flows rapidly, carrying rocks and sediments. As it reaches the plains, the slope becomes gentle and the river slows down. Near its mouth, it deposits sediments and forms fertile land.

52. Why is erosion maximum in the upper course of the river? (1 Marks)

Answer:

Erosion is maximum in the upper course of a river because:

- The river flows down **steep slopes** with great speed and energy.
- The fast-flowing water has a strong capacity to **cut into the river bed and banks**, carrying away rocks and soil.

Therefore, the steep gradient and high velocity make erosion most intense in the upper course.

53. Name the landform formed near the river mouth. (1 Mark)

Answer:

The landform commonly formed near the river mouth is a **delta**.

Note: If a river enters a sea with strong currents or tides, an **estuary** may form instead of a delta. However, in this case, the passage mentions sediment deposition, so **delta** is the correct answer.

54. Why does deposition occur in the lower course? (2 Marks)

Answer:

Deposition occurs in the lower course because:

- The river enters a **plain, lake, or sea**, where the slope becomes gentle.
 - The river's speed decreases, causing it to lose the energy needed to carry sediments.
 - As a result, sand, silt, and clay settle down, forming **deltas, floodplains, and natural levees**.
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Case 2

A glacier slowly moves downhill carrying rocks, sand, and soil. As it moves, it scrapes the land beneath and deposits materials when it melts.

55. Which agent of erosion is described here? (1 Mark)

Answer:

The agent of erosion described is a **glacier**.

A glacier is a large, slow-moving mass of ice that erodes, transports, and deposits rock materials, thereby shaping the Earth's surface.

56. Name one erosional landform formed by glaciers. (1 Mark)

Answer:

One erosional landform formed by glaciers is a **U-shaped valley**.

Other examples: Cirque, Hanging Valley, Fjord.

57. Name one depositional landform formed by glaciers. (1 Mark)

Answer:

One depositional landform formed by glaciers is a **moraine**.

Other examples: Drumlin, Esker, Outwash Plain.
